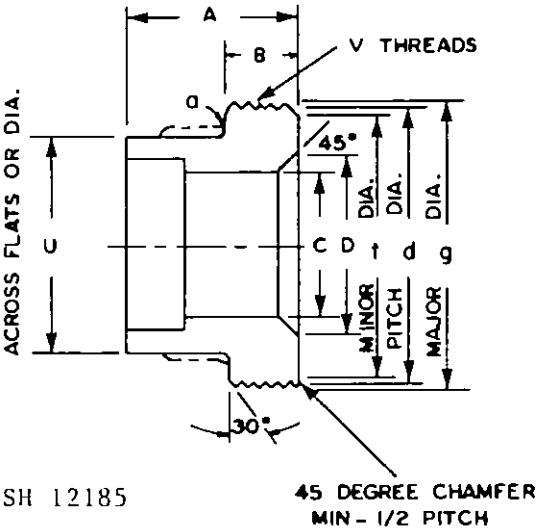
MILITARY SPECIFICATION SHEET

FITTINGS, PIPE, CAST BRONZE, SILVER-BRAZING; UNION, PIPE

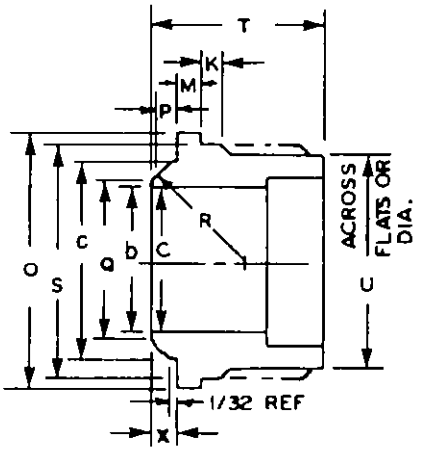
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the unions herein shall consist of this specification and the latest issue of MIL-F-1183.



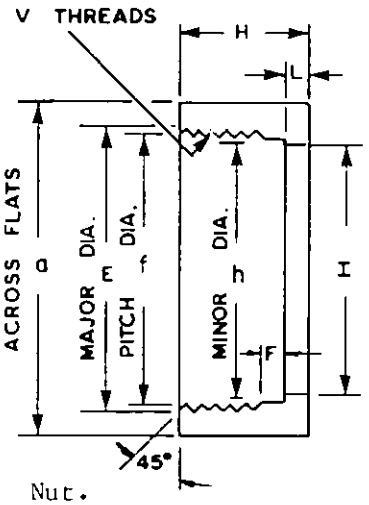
SH 12185

FIGURE 1. Female threadpiece.



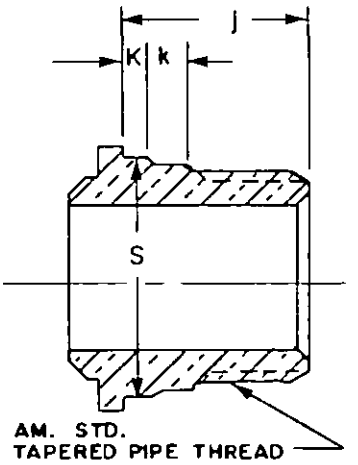
SH 12186

FIGURE 2. Female tailpiece.



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FIGURE 3. Nut.



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FIGURE 4. Male thread tailpiece.

AMSC N/A FSC 4730

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TABLE I. Union components dimensions.

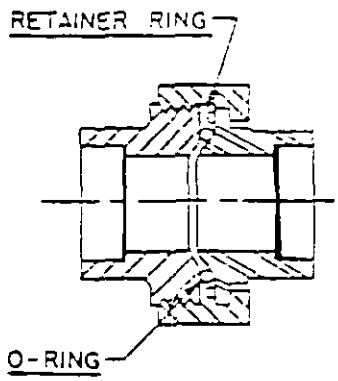
Nominal pipe size	A ± 0.02	B ± 1/32	C Diameter		D Diameter	E	F ± .016	G	H Min	I Diameter	K ± .016	L Min	M ± .010
Inches	Inches	Inch	Inches	±	Inches	Inches	Inch	Inches	Inches	Inches	Inch	Inch	Inch
1/4	0.85	23/64	0.398	0.015	0.610	1.095	0.120	1-11/32	21/32	0.860	0.125	0.125	0.125
3/8	.94	25/64	.532	.015	.755	1.281	.125	1-35/64	45/64	1.020	.135	.135	.135
1/2	.99	25/64	.697	.020	.960	1.491	.135	1-25/32	47/64	1.220	.145	.145	.145
3/4	1.18	1/2	.907	.025	1.140	1.835	.150	2- 3/16	7/8	1.510	.160	.160	.160
1	1.26	1/2	1.171	.025	1.445	2.165	.170	2-35/64	59/64	1.820	.175	.175	.180
1-1/4	1.43	35/64	1.502	.030	1.770	2.601	.200	3- 3/64	1-1/32	2.220	.205	.205	.210
1-1/2	1.52	35/64	1.742	.030	2.060	2.921	.220	3-25/64	1-1/16	2.520	.220	.220	.230
2	1.67	35/64	2.186	.030	2.545	3.521	.250	4- 3/64	1-5/32	3.090	.250	.250	.260

Nominal pipe size	O Diameter	P	Q Diameter	R Radius	S Diameter	T ± .020	U	V	X	Pipe thread size
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches		Inch	Inches
1/4	0.965	0.100	0.510	0.361	0.83	0.874	0.75	16-UNS	0.134	3/8
3/8	1.135	.103	.650	.460	.99	.959	.90	14-UNS	.139	1/2
1/2	1.345	.113	.835	.591	1.19	1.049	1.09	14-UNS	.159	3/4
3/4	1.650	.115	1.030	.728	1.47	1.154	1.35	11-UNS	.154	1
1	1.975	.135	1.315	.930	1.78	1.299	1.63	11-UNS	.184	1-1/4
1-1/4	2.395	.130	1.650	1.167	2.18	1.469	2.00	10-UNS	.174	1-1/2
1-1/2	2.710	.143	1.915	1.355	2.47	1.619	2.27	10-UNS	.199	2
2	3.310	.165	2.395	1.694	3.04	1.854	2.80	10-UNS	.224	2-1/2

Nominal pipe size	a Radius (ref)	b Dia (ref)	c Dia	d Pitch diameter		f Pitch diameter		g Major diameter		h Minor diameter		j	k (ref)	t Minor dia
				Min	Max	Min	Max	Min	Max	Min	Max			Max
Inches	Inch	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches
1/4	0.094	0.441	0.618	1.0456	1.0506	1.0520	1.0585	1.0846	1.0940	1.0278	1.0348	0.985	0.30	1.0133
3/8	.094	.576	.768	1.2250	1.2304	1.2320	1.2390	1.2697	1.2800	1.2043	1.2122	1.155	.30	1.1884
1/2	.094	.741	.972	1.4348	1.4403	1.4420	1.4492	1.4764	1.4867	1.4111	1.4190	1.165	.30	1.3984
3/4	.125	.951	1.174	1.7640	1.7702	1.7720	1.7801	1.8172	1.8293	1.7326	1.7424	1.350	.31	1.7175
1	.125	1.216	1.494	2.0938	2.1001	2.1020	2.1102	2.1471	2.1592	2.0626	2.0724	1.505	.33	2.0475
1-1/4	.125	1.561	1.828	2.5233	2.5301	2.5321	2.5407	2.5821	2.5950	2.4887	2.4992	1.535	.33	2.4723
1-1/2	.125	1.801	2.116	2.8432	2.8501	2.8520	2.8609	2.9021	2.9150	2.8087	2.8192	1.610	.39	2.7923
2	.125	2.276	2.638	3.4430	3.4499	3.4520	3.4610	3.5020	3.5149	3.4087	3.4192	2.020	.39	3.3923

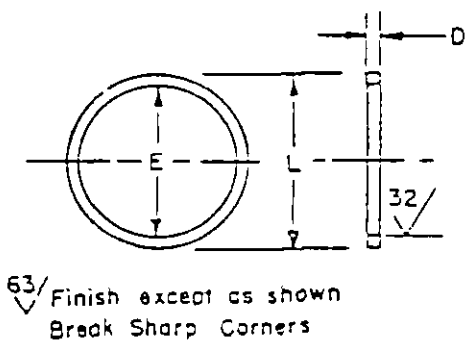
NOTE: For socket details see figure 1 and table V of MIL-F-1183.

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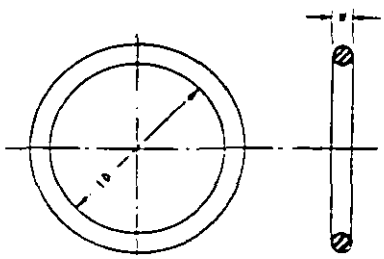
SH 12189

FIGURE 5. O-ring and retainer ring placement.



SH 12190

FIGURE 6. Retainer ring.



SH 12191

FIGURE 7. O-ring.

TABLE II. Retainer ring dimensions.

Nominal pipe size	D ± .005 (Inch)	E ± .005 (Inches)	L ± .005 (Inches)
1/4	0.050	0.813	1.008
3/8	.050	.938	1.184
1/2	.050	1.188	1.394
3/4	.060	1.500	1.715
1	.070	1.750	2.045
1-1/4	.070	2.062	2.471
1-1/2	.070	2.375	2.791
2	.090	3.000	3.391

TABLE III. O-ring configuration and size.

Nominal pipe size	O-ring configuration and size
1/4	AS568-017
3/8	AS568-019
1/2	AS568-023
3/4	AS568-125
1	AS568-129
1-1/4	AS568-134
1-1/2	AS568-139
2	AS568-232

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REQUIREMENTS:

1. Union type. Female unions (socket both ends) shall be type A (streamlined interior). The male tailpiece (see figure 4) shall be type A (streamlined interior).
2. Female union dimensions. Dimensions for the female union components (see figures 1, 2 and 3) shall be as specified in table I. On figure 2, the dimensions P, Q and R define the point of contact of the ground seat.
3. Male tailpiece dimensions. Dimensions for the threaded male tailpiece (see figure 4) shall be as specified in table I. Male tailpiece dimensions are the same as the dimensions for the female tailpiece (see figure 2) except for those dimensions identified in figure 4. NOTE: The size of the male pipe threads is one size larger than the nominal pipe size by which the threaded male tailpiece is identified.
4. Nut dimensions. Dimensions for the union nut (see figure 3) shall be as specified in table I.
5. Static O-ring seal and retainer ring. When specified (see requirement 8 herein), unions shall be provided with an O-ring and a retainer ring which when installed is outboard of the ground joint (see figure 5). The O-ring and retainer do not affect the primary contact of the ground seat and forms an effective secondary seal. NOTE: For Navy applications, unions shall not be used without O-ring and retainer ring.
6. O-ring seal configuration and material. O-ring configuration shall be in accordance with AS 568 of size given in table III for the required union nominal pipe size. O-ring material shall be in accordance with MIL-R-83248 and MIL-R-83248/1, class 1 rubber. For steam service to 300 °F, O-ring material shall be ASTM D 2000, 2AA 815 A13F17EA14Z (300°F steam and phosphate ester base hydraulic fluid service) ethylene propylene rubber.
7. Retainer ring dimensions and material. Retainer ring dimensions shall be as specified in table II. Material of retainer ring shall be QQ-C-390 alloy C90300 or C92200 or ASTM B 61 (UNS no. C92200) bronze.
8. Specification sheet part number. Unions covered by this specification sheet shall be designated in the following form (see 6.2(b) of MIL-F-1183).

MI183/10 - XX X X

Specification sheet code number
Item and pipe size code number (see requirement 9 herein)
Brazing ring groove and brazing ring code letter (see requirement 10 herein)
O-ring and retainer ring code number (see requirement 11 herein)

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9. Item and pipe size. The item (female union, male tailpiece and union nut) and the nominal pipe sizes are identified by two digits (see table IV).

TABLE IV. Code number to item and size.

Female union (figures 1, 2 and 3)	Male tailpiece (figure 4)	Union nut (figure 3)	Nominal pipe size (Inches)
10	30	50	1/4
11	31	51	3/8
12	32	52	1/2
13	33	53	3/4
14	34	54	1
15	35	55	1-1/4
16	36	56	1-1/2
17	37	57	2

10. Brazing ring groove and brazing ring. No groove (face fed, male tailpiece or nut), grooved without brazing ring, grooved with brazing ring inserted in groove, grooved with brazing rings separately packaged and attached to the union or grooved with brazing rings separately packaged and shipped with but not attached to the union socket ends are identified by a single letter (see table V).

TABLE V. Code letter to socket condition.

Code letter	Condition of socket
N	No groove (face fed, male tailpiece or nut)
G 1/	Grooved (no brazing ring)
R 1/	Grooved with brazing ring inserted
A 2/	Grooved with brazing ring separately packaged and attached
S 1/	Grooved with brazing ring separately packaged and shipped with

1/ No National Stock Number (NSN) shall be established or exist in the Federal Supply System for a union with code letter of G, R or S.

2/ When the socket condition code letter is "A", the brazing rings shall not be inserted in the grooves but shall be packaged separately and either included within each individual union package or physically attached to the union.

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11. O-ring seal and retainer ring. Whether the item is provided without O-ring and retainer ring, with O-ring and retainer ring installed, with O-ring and retainer ring separately packaged and attached to the item or with O-ring and retainer ring separately packaged and shipped with the item but not attached to item is identified by a digit (see table VI).

TABLE VI. Code number to O-ring and retainer ring requirement.

Code number	O-ring and retainer ring requirement
1	No O-ring and retainer ring (for male tailpiece and union nut only)
2/ 2	Without O-ring and retainer ring (for female union only)
3	With O-ring (air, oil or water service) and retainer ring installed 1/
4	With O-ring (steam service) and retainer ring installed 1/
2/ 5	With O-ring (air, oil or water service) and retainer ring separately packaged and attached
2/ 6	With O-ring (steam service) and retainer ring separately packaged and attached
2/ 7	With O-ring (air, oil and water service) and retainer ring separately packaged and shipped with
2/ 8	With O-ring (steam service) and retainer ring separately packaged and shipped with

1/ This requirement is not applicable to male tailpiece (see figure 4) or union nut (see figure 3).

2/ No National Stock Number (NSN) shall be established or exist in the Federal Supply System for a female union with code number of 1, 2, 5, 6, 7 or 8 and for a male tailpiece or union nut with code number 2, 3, 4, 5, 6, 7 or 8.

12. Specification part number definition. The specification sheet part number is a definitive part number which is formed by combining the specification sheet code number (M1183/10 for MIL-F-1183/10) with a dash after it, the item and pipe size code number, the brazing ring groove and brazing ring code letter and the O-ring and retainer ring code number as specified in requirement 8.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - SH
Air Force - 99

Review activity:
DLA - CS

Preparing activity:
Navy - SH

Agent:
DLA - CS

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